Worksheet 1-5 A
Limit Properties & Piecewise Functions #1
AP Calculus Name_____
BE SURE TO SHOW ALL WORK.
Use the given expressions to find the limits:

$\lim_{x\to 2} f(x) = -1$ $\lim_{x\to 1} f(x) = 9$ $\lim_{x\to 3} f(x) = 4$

$\lim_{x\to 4} f(x) = 2$ $\lim_{x\to 3} g(x) = 1$

1.) $\lim_{x\to 3}[2f(x) + g(x)] =$

2.) $\lim_{x\to 3}\left[\frac{f(x)}{g(x)} + 2x^2\right] =$

3.) $\lim_{x\to 1}\left[4\sqrt{f(x)}\right] =$

4.) $\lim_{x\to 2}\left[f(2x) + f\left(\frac{x}{2}\right)\right] =$

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5.) Use the table to find the given limits

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|---|---|
| $f(-2) = 7$ | $\lim_{x\to -2} f(x) = 2$ |
| $g(-2) = 1$ | $\lim_{x\to -2} g(x) = -1$ |
| $h(-2) = -4$ | $\lim_{x\to -2} h(x) = -3$ |

a.) $\lim_{x\to -2}[h(x)(2f(x))] + h(-2)$

b.) $f(-2)\lim_{x\to -2}[g(x) - h(x)]$

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$$f(x) = \begin{cases} x^2 & 0 \le x \le 2 \\ \frac{-2}{3}(x-2) + 2 & 2 < x < 5 \\ \frac{3}{2}(x-5) & 5 \le x \le 7 \end{cases}$$

6.) The function $f(x)$ is defined above. Evaluate the following limits or explain why the limit does not exist.

a.) $\lim_{x\to 2^-} f(x)$

b.) $\lim_{x\to 2^+} f(x)$

c.) $\lim_{x\to 2} f(x)$

$$h(x) = \begin{cases} x^2 - 1 & x < 2 \\ 4x - 5 & x \geq 2 \end{cases}$$

7.) The function $g(x)$ is defined above. Evaluate the following limits or explain why the limit does not exist.

a.) $\lim_{x \to 2^-} h(x)$ b.) $\lim_{x \to 2^+} h(x)$ c.) $\lim_{x \to 2} h(x)$ d.) $h(0)$

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$$g(x) = \begin{cases} x^2 + x - 1 & x \leq 0 \\ 2x + 4 & x > 0 \end{cases}$$

8.) The function $g(x)$ is defined above. Evaluate the following limits or explain why the limit does not exist.

a.) $\lim_{x \to 0^-} g(x)$ b.) $\lim_{x \to 0^+} g(x)$ c.) $\lim_{x \to 0} g(x)$ d.) $g(0)$

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$$p(x) = \begin{cases} \cos(x) & x \leq 0 \\ x^2 & 0 < x < 3 \\ 10 - x & x \geq 3 \end{cases}$$

9.) The function $p(x)$ is defined above. Evaluate the following limits or explain why the limit does not exist.

a.) $\lim_{x \to 0} p(x)$ b.) $\lim_{x \to 3} p(x)$

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$$k(x) = \begin{cases} ax - 4 & x < 2 \\ x^3 - 2ax & x \geq 2 \end{cases}$$

10.) Find the value of "$a$" in the piecewise function so that $\lim_{x \to 2} k(x)$ exists.